Solar System Models · TA Brief

Learning goal: I can use solar system data and a scale model to link distance from the Sun to orbit time.

Watch (staff preview first)

Short silent teacher demonstration clip for this lesson's model and practical sequence · preview before class use.

Adult Support Routine

No cold calling; nobody is asked to read difficult text aloud.

Give time to start -> point to the relevant place -> rephrase -> offer a small prompt or word bank -> reduce to a limited choice.

Beside the You do in the pupil book, mark I (independent), S (supported) or C (choice).

Must-Know Words

orbit time: the time a planet takes to travel once all the way around the Sun (its year)

scale model: a model that shrinks huge real distances or sizes using a fixed rule

astronomical data: measurements of planets such as distance from the Sun, diameter, and orbit time

limitation: something a science model leaves out or cannot show accurately

Key Answers

A. Planets further from the Sun travel a much longer path and move more slowly in orbit.

A. Following the pattern, Saturn's orbit time (29 Earth years) sits between Jupiter (12 years) and Uranus (84 years).

1. Neptune (165 Earth years). 2. As distance from the Sun increases, the time taken for one complete orbit increases. 3. Strength: shows the true relative spacing (inner 4 bunched, outer 4 wide apart). Limitation: planets are shown stationary in a line, and planet dot sizes are not to the same scale.